

MATHEMATICS AND STATISTICS

TIME: 3 HOURS

[MAX. MARKS: 80]

Note: (i) All questions are compulsory.

(ii) Figures to the right indicate full marks.

(iii) Graph of L.P.P should be drawn on graph paper only.

(iv) Use of logarithmic table is allowed.

(v) Answers to the questions of Section – I and Section –II should be written in the only one answer book.

(vi) Answer to every new question must be written on a new page.

Question 1

[12]

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-equation: [6]

(i) The negation of $p \wedge (q \rightarrow r)$ is..... (2)

(a) $p \vee (\sim q \vee r)$

(b) $\sim p \wedge (q \rightarrow r)$

(c) $\sim p \wedge (\sim q \rightarrow \sim r)$

(d) $\sim p \vee (q \wedge \sim r)$

Sol:

The negation of $p \wedge (q \rightarrow r)$ is

$$\sim [p \vee (q \rightarrow r)]$$

$$\equiv \sim p \vee \sim (q \rightarrow r)$$

$$\equiv \sim p \vee (q \wedge \sim r)$$

Hence, the correct option is (c).

(ii) If $\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$ then x is.....

(2)

(a) $\frac{-1}{2}$

(b) 1

(c) 0

(d) $\frac{1}{2}$

Sol:

We have,

$$\sin^{-1}(1-x) - 2\sin^{-1}x = \frac{\pi}{2}$$

$$\sin^{-1}(1-x) = \frac{\pi}{2} + 2\sin^{-1}x$$

$$1-x = \sin\left(\frac{\pi}{2} + 2\sin^{-1}x\right)$$

$$1-x = \cos(2\sin^{-1}x)$$

$$1-x = \cos[\cos^{-1}(1-2x^2)]$$

$$1-x = 1-2x^2$$

$$2x^2 - x = 0$$

$$x(2x-1) = 0$$

$$\therefore x = 0 \text{ or } x = \frac{1}{2}$$

For,

$$x = \frac{1}{2}$$

$$\sin^{-1}(1-x) - 2\sin^{-1}x = \sin^{-1}(1) - 2\sin^{-1}(0) = \frac{\pi}{2} - 0 = \frac{\pi}{2}$$

Hence, the correct option is (c).

(iii) The joint equation of the pair of lines passing through (2, 3) and parallel to the coordinate axes is.....

(2)

(a) $xy - 3x - 2y + 6 = 0$

(b) $xy + 3x + 2y + 6 = 0$

(c) $xy = 0$

(d) $xy - 3x - 2y - 6 = 0$

Sol:

We have,

Equation of the coordinate axes are $x=0$ and $y=0$

Therefore, the equations of the lines passing through (2, 3) and parallel to coordinate axes are,

$$x=2 \text{ and } y=3$$

$$x-2=0 \text{ and } y-3=0$$

The joint equation is given a

$$(x-2)(y-3)=0$$

$$xy-3x-2y+6=0$$

Hence, the correct option is (a).

(B) Attempt any THREE of the following:

[6]

(i) Find $(AB)^{-1}$ if $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 1 & -2 \end{bmatrix}$. **(2)**

Sol:

Given that,

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 1 & -2 \end{bmatrix}$$

now,

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 1 & -2 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1+2+3 & -1+4-6 \\ 1-2-3 & -1-4+6 \end{bmatrix}$$

$$\therefore AB = \begin{bmatrix} 6 & -3 \\ -4 & 1 \end{bmatrix}$$

Now, we get

$$M_{11} = 1, \quad A_{11} = (-1)^{1+1}(1) = 1$$

$$M_{12} = -4, \quad A_{12} = (-1)^{1+2}(-4) = 4$$

$$M_{21} = -3, \quad A_{21} = (-1)^{2+1}(-3) = 3$$

$$M_{22} = 6, \quad A_{22} = (-1)^{2+2}(6) = 6$$

$$\therefore \text{adj}(AB) = \begin{bmatrix} 1 & 4 \\ 3 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$$

and,

$$|AB| = \begin{vmatrix} 1 & 3 \\ 4 & 6 \end{vmatrix} = 6 - 12 = -6 \neq 0$$

Therefore using,

$$A^{-1} = \frac{1}{|A|} \text{adj}A$$

$$\therefore (AB)^{-1} = \frac{-1}{6} \begin{bmatrix} 1 & 3 \\ 4 & 6 \end{bmatrix}$$

Hence, the required inverse.

(ii) Find the vector equation of the plane passing through a point having position vector $3\hat{i} - 2\hat{j} + \hat{k}$ and perpendicular to the vector $4\hat{i} + 3\hat{j} + 2\hat{k}$.

(2)

Sol:

We know that the equation of the plane passing through a point $A(\vec{a})$ and normal to $\vec{n}$ is
 $\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$

Here,

$$\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k} \quad \text{and} \quad \vec{n} = 4\hat{i} + 3\hat{j} + 2\hat{k}$$

Therefore, the equation of the required plane is

$$\vec{r} \cdot (4\hat{i} + 3\hat{j} + 2\hat{k}) = (3\hat{i} - 2\hat{j} + \hat{k}) \cdot (4\hat{i} + 3\hat{j} + 2\hat{k})$$

$$\vec{r} \cdot (4\hat{i} + 3\hat{j} + 2\hat{k}) = 12 - 6 + 2$$

$$\vec{r} \cdot (4\hat{i} + 3\hat{j} + 2\hat{k}) = 8$$

Hence, this is the required vector equation.

(iii) If $\vec{p} = \hat{i} - 2\hat{j} + \hat{k}$ and $\vec{q} = \hat{i} + 4\hat{j} - 2\hat{k}$ are position vector (P.V) of points P and Q, find the position vector of the point R which divides segment PQ internally in the ratio 2:1.

(2)

Sol:

Let $R(\vec{r})$ is the point which divides the line segment joining the points p and Q internally in the ratio 2:1.

$$\vec{r} = \frac{2(\hat{i} + 4\hat{j} - 2\hat{k}) + 1(\hat{i} - 2\hat{j} + \hat{k})}{2+1}$$

$$\vec{r} = \frac{3\hat{i} + 6\hat{j} - 3\hat{k}}{3}$$

$$\therefore \vec{r} = \hat{i} + 2\hat{j} - \hat{k}$$

And the coordinates of the point R are (1, 2, -1)

(iv) Find K, if one of the lines given by $6x^2 + kxy + y^2 = 0$ is $2x + y = 0$. (2)

Sol:

Let m_1 be the slope of $2x + y = 0$

$$\therefore m_1 = -2$$

Now, comparing $6x^2 + kxy + y^2 = 0$ with $ax^2 + 2hxy + by^2 = 0$

We get,

$$a = 6, h = \frac{k}{2}, b = 1$$

$$\therefore m_1 + m_2 = \frac{-2h}{b} = -k$$

$$\therefore -2 + m_2 = k$$

$$\Rightarrow m_2 = 2 - k$$

Now,

$$m_1 \cdot m_2 = \frac{a}{b}$$

$$(-2)(2 - k) = \frac{6}{1}$$

$$-4 + 2k = 6$$

$$k = 5$$

(v) If the lines $\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$ and $\frac{x-1}{3k} = \frac{y-5}{1} = \frac{z-6}{-5}$ are at right angle then find the value of K. (2)

Sol;

We have,

$$\frac{x-1}{-3} = \frac{y-2}{2k} = \frac{z-3}{2}$$

and,

$$\frac{x-1}{3k} = \frac{y-5}{1} = \frac{z-6}{-5}$$

Now, direction ratios of line (1) are $-3, 2k$ and line (2) are $3k, 1, -5$.

If lines are perpendicular, then

$$(-3)(3k) + (2k)(1) + (2)(-5) = 0$$

$$-9k + 2k - 10 = 0$$

$$-7k = 10$$

$$k = -\frac{10}{7}$$

Question 2

[14]

(A) Attempt any TWO of the following:

[6]

(i) Examine whether the following logical statement pattern is tautology, contradiction or contingency.

$$[(p \rightarrow q) \wedge q] \rightarrow p$$

(3)

Sol:

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge q$	$[(p \rightarrow q) \wedge q] \rightarrow p$
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	T

Since the truth table contains neither all T nor F in the last column.

$\therefore [(p \rightarrow q) \wedge q] \rightarrow p$ is a contingency.

(ii) By vector method prove that the medians of a triangle are concurrent.

(3)

Sol:

1	2	3	4	5	6	7	8
P	q	$p \leftrightarrow q$	$\sim p$	$\sim q$	$p \wedge q$	$\sim p \wedge \sim q$	$(p \wedge q) \vee (\sim p \wedge \sim q)$
T	T	T	F	F	T	F	T
T	F	F	F	T	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	T	F	T	T

The entries in (3) and (8) column are identical.

$$\therefore p \leftrightarrow q = (p \wedge q) \vee (\sim p \wedge \sim q)$$

(iii) Find the shortest distance between the lines $\vec{r} = (4\hat{i} - \hat{j}) + \lambda(\hat{i} + 2\hat{j} - 3\hat{k})$ and $\vec{r} = (\hat{i} - \hat{j} + 2\hat{k}) + \mu(\hat{i} + 4\hat{j} - 5\hat{k})$ Where λ and μ are parameters. (3)

Sol:

We know that the shortest distance between the lines, $\vec{r} = \vec{a}_1 + \lambda \vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu \vec{b}_2$ is given as,

$$d = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$$

Here,

$$\vec{a}_1 = 4\hat{i} - \hat{j}, \vec{a}_2 = \hat{i} - \hat{j} + 2\hat{k}, \vec{b}_1 = \hat{i} + 2\hat{j} - 3\hat{k}, \vec{b}_2 = \hat{i} + 4\hat{j} - 5\hat{k}$$

Now,

$$\begin{aligned} \vec{a}_2 - \vec{a}_1 &= (\hat{i} - \hat{j} + 2\hat{k}) - (4\hat{i} - \hat{j}) = -3\hat{i} + 0\hat{j} + 2\hat{k} \\ &= -3\hat{i} + 2\hat{k} \end{aligned}$$

And,

$$\begin{aligned} \vec{b}_1 \times \vec{b}_2 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix} \\ &= \hat{i}(-10 + 12) - \hat{j}(-5 + 3) + \hat{k}(4 - 2) \\ &= 2\hat{i} + 2\hat{j} + 2\hat{k} \\ \therefore (\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2) &= (-3\hat{i} + 2\hat{k}) \cdot (2\hat{i} + 2\hat{j} + 2\hat{k}) = -6 + 4 = -2 \end{aligned}$$

And,

$$|\vec{b}_1 \times \vec{b}_2| = \sqrt{(2)^2 + (2)^2 + (2)^2} = \sqrt{12}$$

$\therefore$ Shortest distance,

$$\begin{aligned} d &= \left| \frac{-2}{\sqrt{12}} \right| \\ &= \frac{2}{2\sqrt{3}} = \frac{1}{\sqrt{3}} \text{ units.} \end{aligned}$$

(B) Attempt any TWO of the following:

[8]

(i) In ΔABC with the usual notations prove that

$$(a-b)^2 \cos^2 \left(\frac{C}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) = c^2 \quad (4)$$

Sol:

By the projection rule in ΔABC

$$a = c \cos B + b \cos C$$

$$b = a \cos C + c \cos A$$

$$a + b = c \cos B + b \cos C + a \cos C + c \cos A$$

$$(a+b)(1 - \cos C) = c(\cos B + \cos A)$$

$$(a+b) \left(2 \sin^2 \left(\frac{C}{2} \right) \right) = c(\cos B + \cos A)$$

$$(a+b)^2 \cdot \sin^2 \left(\frac{C}{2} \right) = \frac{c}{2} (\cos B + \cos A)(a+b) \dots \dots \dots (1)$$

And,

$$a - b = c \cos B + b \cos C - a \cos C - c \cos A$$

$$(a-b)(1 + \cos C) = c(\cos B - \cos A)$$

$$(a-b) \left(\cos^2 \left(\frac{C}{2} \right) \right) = c(\cos B - \cos A)$$

$$(a-b)^2 \cos^2 \left(\frac{C}{2} \right) = \frac{c}{2} (\cos B - \cos A)(a-b) \dots \dots \dots (2)$$

Now, adding the equations (1) and (2), we get

$$\begin{aligned} & (a-b)^2 \cos^2 \left(\frac{C}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) \\ &= \frac{c}{2} [a \cos B + a \cos A + b \cos B + b \cos A + a \cos B - a \cos A - b \cos B + b \cos A] \\ &\therefore (a-b)^2 \cos^2 \left(\frac{C}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) = \frac{c}{2} 2(a \cos B + b \cos A) \\ &\therefore (a-b)^2 \cos^2 \left(\frac{C}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) = c^2 \quad [\because c = a \cos B + b \cos A] \end{aligned}$$

(ii) Minimize $Z = 4x + 5y$ subject to $2x + y \geq 7, 2x + 3y \leq 15, x \leq 3, x \geq 0, y \geq 0$,solve using graphical method. (4)

Sol:

Corner Point Method:

Step:1

Convert the in equations into the equations and draw the corresponding line.

$$2x + y = 7 \Rightarrow \frac{x}{7/2} + \frac{y}{7} = 1$$

$$2x + 3y = 15 \Rightarrow \frac{x}{15/2} + \frac{y}{5} = 1$$

$$x = 3$$

and $x \geq 0, y \geq 0$, the solution lies in the first quadrant.

$\therefore$ ABCA is the feasible region.

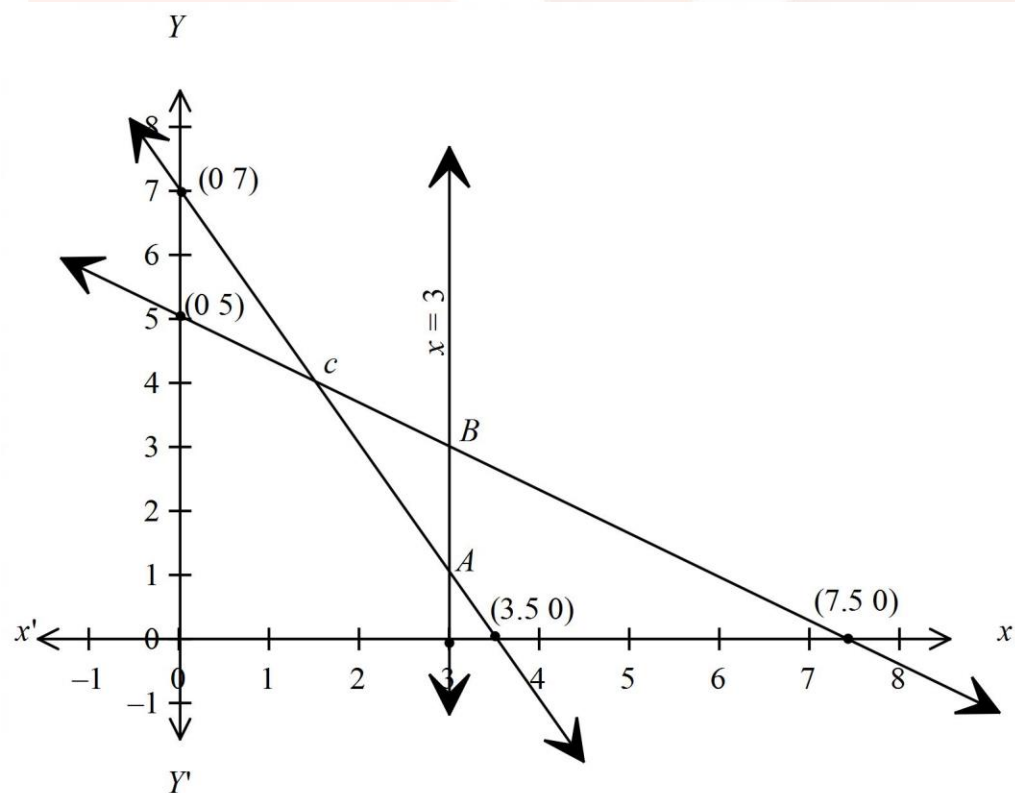
Step2:

We find the coordinates of the vertices A,B,C

$$A = (3, 1)$$

$$B = (3, 3)$$

$$C = \left(\frac{3}{2}, 4\right)$$



Step 3:

We find the values of the objective function, $Z = 4x + 5y$ at the corner points (vertices) A, B and C respectively.

$$Z = (A) = 4(3) + 5(1) = 12 + 5 = 17$$

$$Z = (B) = 4(3) + 5(3) = 12 + 15 = 27$$

$$Z = 4\left(\frac{3}{2}\right) + 5(4) = 6 + 20 = 26$$

$\therefore Z$ has minimum value 17 at (3,1).

$\therefore$ Optimum solution is $x = 3, y = 1$ with minimum value.

(iii) The cost of 4 dozen pencils, 3 dozen pens and 2 dozen erasers is ₹60. The cost of 2 dozen pencils, 4 dozen pens and 6 dozen erasers is ₹90 whereas the cost of 6 dozen pencils, 2 dozen pens and 3 dozen erasers is ₹70. Find the cost of each items per dozen by using matrices.
(4)

Sol:

Let the cost of 1 dozen pencils, 1 dozen pens and 1 dozen of erasers be ₹ x , ₹ y and ₹ z respectively.

$\therefore$ From the given information,

$$4x + 3y + 2z = 60$$

$$2x + 4y + 6z = 90 \text{ i.e } x + 2y + 3z = 45$$

$$6x + 2y + 3z = 70$$

The equation can be written in the matrix form as:

$$\begin{bmatrix} 4 & 3 & 2 \\ 1 & 2 & 3 \\ 6 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 60 \\ 45 \\ 70 \end{bmatrix}$$

On applying $R_1 \leftrightarrow R_2$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 3 & 2 \\ 6 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 45 \\ 60 \\ 70 \end{bmatrix}$$

On applying $R_2 \rightarrow R_2 - 4R_1$ and $R_3 \rightarrow R_3 - 6R_1$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & -10 & -15 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 45 \\ -120 \\ -200 \end{bmatrix}$$

On applying $R_3 \rightarrow R_3 - 2R_2$, we get

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -10 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 45 \\ -120 \\ 40 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x+2y+3z \\ -5y-10z \\ 5z \end{bmatrix} = \begin{bmatrix} 45 \\ -120 \\ 40 \end{bmatrix}$$

By equality of matrices, we get

$$x + 2y + 3z = 45 \dots\dots\dots(1)$$

$$-5y - 10z = -120 \dots\dots\dots(2)$$

$$5z = 40 \dots\dots\dots(3)$$

From equation(3)

$$z = 8$$

On substituting $z = 8$ in equation (2), we get

$$-5y - 80 = -120$$

$$-5y = -40$$

$$\therefore y = 8$$

On substituting $y = 8$ and $z = 8$ in equation (1), we get

$$x + 16 + 24 = 45$$

$$x + 40 = 45$$

$$x = 45 - 40$$

$$x = 5$$

$$\therefore x = 5, y = 8, z = 8$$

$\therefore$ cost of a dozen pencils is ₹5, of a dozen of pens is ₹8 and of a dozen of erasers is ₹8.

Question 3

[14]

(A) Attempt any TWO of the following:

[6]

(i) find the volume of tetrahedron whose coterminous edges are $7\hat{i} + \hat{k}$; $2\hat{i} + 5\hat{j} - 3\hat{k}$ and $4\hat{i} + 3\hat{j} + \hat{k}$. (3)

Sol:

We know that the volume of a tetrahedron with coterminous edges $\vec{a}, \vec{b}$ and $\vec{c}$ is $\frac{1}{6}[\vec{a}, \vec{b}, \vec{c}]$

Here,

$$a = 7\hat{i} + \hat{k}, b = 2\hat{i} + 5\hat{j} - 3\hat{k} \text{ and } c = 4\hat{i} + 3\hat{j} + \hat{k}$$

$$[\vec{a}, \vec{b}, \vec{c}] = \vec{a} \cdot (\vec{b} \times \vec{c})$$

$$= \begin{vmatrix} 7 & 0 & 1 \\ 2 & 5 & -3 \\ 4 & 3 & 1 \end{vmatrix}$$

$$= 7(5+9) - 0(2+12) + 1(6-20)$$

$$= 98 - 0 - 14 = 84$$

$$\text{Volume of tetrahedron} = \frac{1}{6}[\vec{a}, \vec{b}, \vec{c}] = \frac{1}{6} \times 84 = 14 \text{ cubic units.}$$

(ii) Without using truth table show that

$$\sim (p \vee q) \vee (\sim p \wedge q) \equiv \sim p \quad (3)$$

Sol:

Given that,

$$L.H.S = \sim (p \vee q) \vee (\sim p \wedge q)$$

$$= \sim (\sim p \wedge \sim q) \vee (\sim p \wedge q) \quad (\text{By De Morgan's law})$$

$$= \sim p \wedge (\sim q \vee q) \quad (\text{By Distributive law})$$

$$= \sim p \wedge T \quad (\text{By Complement law})$$

$$= \sim p$$

Hence, the RHS is proved.

(iii) Show that every homogeneous equations of degree two in x and y i.e,

$$ax^2 + 2hxy + by^2 = 0 \text{ represents a pair of lines passing through origin if } h^2 - ab \geq 0$$

(3)

Sol:

Let us consider a homogeneous equation of degree two in x and y.

Then,

$$ax^2 + 2hxy + by^2 = 0 \text{-----(1)}$$

Where, either a or b or h is non-zero.

We consider two case:

Case 1

If $b = 0$, then equation becomes

$$ax^2 + 2hxy = 0$$

$$x(ax + 2hy) = 0$$

Which is a joint equation of lines $x = 0$ and $ax + 2hy = 0$.

Thus, the lines pass through the origin.

Case 2:

If $b \neq 0$

Multiplying both sides in equation (1) by b, we get

$$abx^2 + 2hbxy + b^2y^2 = 0$$

$$b^2y^2 + 2hbxy = -abx^2$$

$$b^2y^2 + 2hbxy + h^2x^2 = -abx^2 + h^2x^2$$

$$(by + hx)^2 = \left[\left(\sqrt{h^2 - ab} \right) x \right]^2$$

$$(by + hx)^2 - \left[\left(\sqrt{h^2 - ab} \right) x \right]^2 = 0$$

$$\left[(by + hx) + \left(\sqrt{h^2 - ab} \right) x \right] \left[(by - hx) - \left(\sqrt{h^2 - ab} \right) x \right] = 0$$

Now, joint equation of two lines,

$$(by + hx) + \left(\sqrt{h^2 - ab} \right) x = 0 \text{ and } (by + hx) - \left(\sqrt{h^2 - ab} \right) x = 0$$

$$\left(h + \sqrt{h^2 - ab} \right) x + by = 0 \text{ and } \left(h - \sqrt{h^2 - ab} \right) x + by = 0$$

These lines pass through the origin when $h^2 - ab \geq 0$.

Hence, the equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines passing through the origin.

(B) Attempt any TWO of the following:

[8]

(i) If a line is drawn from the point A(1, 2, 1) is perpendicular to the line joining P(1, 4, 6) and Q(5, 4, 4) then find the coordinates of the foot of the perpendicular.

(4)

Sol:

The equation of a line joining the points P(1, 4, 6) and Q(5, 4, 4) is

$$\frac{x-1}{4} = \frac{y-4}{4-4} = \frac{z-6}{4-6}$$

$$\frac{x-1}{4} = \frac{y-4}{0} = \frac{z-6}{-2}$$

Now, let M be the foot of the perpendicular drawn from the point A(1, 2, 1).

Coordinates of M are given by,

$$\frac{x-1}{4} = \frac{y-4}{0} = \frac{z-6}{-2}$$

$$x = 4\lambda + 1, y = 0\lambda + 4, z = -2\lambda + 6$$

$$\therefore M \equiv (4\lambda + 1, 4, -2\lambda + 6)$$

So, the direction ratios of given line are 4, 0, -2, Since AM is perpendicular to the given line

$$\therefore 4(4\lambda) + 0(2) + (-2)(-2\lambda + 5) = 0$$

$$16\lambda + 4\lambda - 10 = 0$$

$$\lambda = \frac{1}{2}$$

Putting $\lambda = \frac{1}{2}$ in equation (2), we get

$$M \equiv (3, 4, 5)$$

Hence, the Co-ordinates of the foot of the perpendicular are (3, 4, 5)

(ii) Find the vector equation of the plane passing through the point

$\hat{i} + \hat{j} - 2\hat{k}, \hat{i} + 2\hat{j} + \hat{k}, 2\hat{i} - \hat{j} + \hat{k}$. Hence find the cartesian equation of the plane.

(4)

Sol:

$\vec{a}, \vec{b}, \vec{c}$ are the position vectors

$$\text{i.e. } \vec{a} = \hat{i} + \hat{j} - 2\hat{k}, \vec{b} = \hat{i} + 2\hat{j} + \hat{k}, \vec{c} = 2\hat{i} - \hat{j} + \hat{k}$$

and,

$$\vec{b} - \vec{c} = 0\hat{i} + \hat{j} + 3\hat{k}$$

$$\vec{c} - \vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$$

Now,

$$(\vec{b} - \vec{c}) \times (\vec{c} - \vec{a}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 3 \\ 1 & -2 & 3 \end{vmatrix}$$

$$= \hat{i}(3+6) - \hat{j}(0-3) + \hat{k}(0-1)$$

$$= 9\hat{i} + 3\hat{j} - \hat{k}$$

Now, let

$$\vec{n} = (\vec{b} - \vec{c}) \times (\vec{c} - \vec{a}) = 9\hat{i} + 3\hat{j} - \hat{k}$$

Then, the vector equation of the required plane is

$$\vec{r} \cdot \vec{n} = \vec{a} \cdot \vec{n}$$

$$\vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) = (\hat{i} + \hat{j} - 2\hat{k})(9\hat{i} + 3\hat{j} - \hat{k})$$

$$\vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) = 9 + 3 + 2$$

$$\vec{r}(9\hat{i} + 3\hat{j} - \hat{k}) = 14$$

The Cartesian equation of the plane is given by

$$(x\hat{i} + y\hat{j} + z\hat{k})(9\hat{i} + 3\hat{j} - \hat{k}) = 14$$

$$\therefore [\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}]$$

$$9x + 3y - z = 14$$

(iv) Find the general solution of $\sin x + \sin 3x + \sin 5x = 0$ (4)

Sol:

We have,

$$\sin x + \sin 3x + \sin 5x = 0$$

$$(\sin x + \sin 5x) + \sin 3x = 0$$

$$2 \sin \left(\frac{x+5x}{2} \right) \cos \left(\frac{5x-x}{2} \right) + \sin 3x = 0$$

$$2 \sin 3x \cos 2x + \sin 3x = 0$$

$$\sin 3x(2 \cos 2x + 1) = 0$$

Either, $\sin 3x = 0$ or $2\cos 2x + 1 = 0$

$$\text{i.e. } \sin 3x = 0 \text{ or } \cos 2x = -\frac{1}{2}$$

now,

$$\cos 2x = -\cos \frac{\pi}{3}$$

$$\cos 2x = \cos \left(\pi - \frac{\pi}{3} \right)$$

$$\cos 2x = \cos \frac{2\pi}{3}$$

$$\therefore \sin 3x = 0 \text{ or } \cos 2x = \cos \frac{2\pi}{3}$$

$$3x = n\pi, n \in \mathbb{Z} \text{ or } 2x = 2m\pi \pm \frac{2\pi}{3} \text{ where } m \in \mathbb{Z}$$

$$\text{Hence, } x = \frac{n\pi}{3}, n \in \mathbb{Z} \text{ or } x = m\pi \pm \frac{\pi}{3}, \text{ where } n, m \in \mathbb{Z}$$

SECTION-II

Question 4

[12]

(A) Select and write the appropriate answer from the given alternatives in each of the following sub-questions:

[6]

(i) If the function

$$f(x) = k + x, \text{ for } x < 1$$

$$= 4x + 3, \text{ for } x \geq 1$$

Is continuous at $x=1$ then $K = \dots\dots\dots$

(2)

(a) 7

(b) 8

(c) 6

(d) -6

Sol:

$f(x)$ is continuous at $x=1$

$$\therefore \lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^+} f(x) = f(1)$$

Hence,

$$\lim_{x \rightarrow 1^-} f(k+x) = \lim_{x \rightarrow 1^+} (4x+3)$$

$$k+1 = 4(1)+3$$

$$k = 7-1$$

$$k = 6$$

Hence, the correct options is (c)

(ii) The equation of tangent to the curve $y = x^2 + 4x + 1$ at $(-1, -2)$ is.....

(2)

(a) $2x - y = 0$

(b) $2x + y - 5 = 0$

(b) $2x - y - 1 = 0$

(d) $x + y - 1 = 0$

Sol:

Let us consider the given equation of the curve is,

$$y = x^2 + 4x + 1$$

On differentiating w.r.t.x, we get,

$$\frac{dy}{dx} = 2x + 4$$

$$\therefore \left(\frac{dy}{dx} \right)_{x=-1} = 2(-1) + 4 = -2 + 4 = 2$$

$\therefore$ Slope of tangent at $(-1, -2)$ is 2.

The equation of the tangent at $(-1, -2)$ is

$$y - (-2) = 2(x - (-1))$$

$$y + 2 = 2(x + 1)$$

$$y + 2 = 2x + 2$$

$$2x - y = 0$$

Hence, the correct option is (a).

(iii) Given that $2x - y = 0$ then the value of p is.....

(2)

(a) 0.6

(b) 0.7

(c) 0.8

(d) 0.4

Sol:

Given that,

$$X \sim B(n=10, P)$$

$$E(X) = 8$$

$$E(X) = nP$$

$$\therefore nP = 8$$

$$10P = 8$$

$$P = 0.8$$

Hence, the correct option is (c).

(B) Attempt any THREE of the following:

[6]

(i) if $x + y - 1 = 0$ find $x + y - 1 = 0$

(2)

Sol;

We have,

$$y = x^x$$

$$\therefore \log y = x \log x$$

On differentiating w.r.t.x, we get,

$$\frac{1}{y} \frac{dy}{dx} = \frac{x}{x} + \log x$$

$$\frac{dy}{dx} = y + y \log x$$

$$\frac{dy}{dx} = x^x (1 + \log x) \quad (\because y = x^x)$$

(ii) The displacement 's' of a moving particle at time 't' is given by $x + y - 1 = 0$. Find its acceleration when the velocity is zero.

(2)

Sol:

Given that,

$$s = 5 + 20t - 2t^2$$

$$\text{velocity}(v) = \frac{ds}{dt} = 20 - 4t$$

Now,

$$v = 0$$

$$\therefore 20 - 4t = 0$$

$$\therefore t = 5$$

And acceleration(a)

$$\frac{dv}{dt} = -4$$

$$\therefore \left(\frac{dv}{dt} \right)_{t=5} = -4$$

Hence, the acceleration is -4 unit/sec^2 .

(iii) Find the area bounded by the curve $x + y - 1 = 0$, x-axis and the lines $x + y - 1 = 0$ and $x + y - 1 = 0$

(2)

Sol:

Let the required area,

(Figher)

$$\begin{aligned} (A) &= \int_0^a y dx \\ &= \int_0^a 2\sqrt{ax} dx \\ &= 2\sqrt{a} \cdot \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^a \\ &= 2\sqrt{a} \cdot \frac{2}{3} \left[x^{\frac{3}{2}} \right]_0^a \\ &= \frac{4}{3} \sqrt{a} \left(a^{\frac{3}{2}} - 0 \right) \\ &= \frac{4}{3} a^2 \text{ sq. units} \end{aligned}$$

(iv) The probability distribution of a discrete random variable X is:

(2)

$X = x$	1	2	3	4	5
$P(X = x)$	K	2k	3k	4k	5k

Find $P(X \leq 4)$.

Sol:

We have,

$$k + 2k + 3k + 4k + 5k = 1$$

$$15k = 1$$

$$k = \frac{1}{15}$$

Now,

$$P(X \leq 4) = k + 2k + 3k + 4k$$

$$= \frac{1}{15} + \frac{2}{15} + \frac{3}{15} + \frac{4}{15}$$

$$= \frac{10}{15}$$

$$\therefore P(X \leq 4) = \frac{2}{3}$$

(v) Evaluate: $\int \frac{\sin x}{\sqrt{36 - \cos^2 x}} dx$ (2)

Sol;

We have,

$$\int \frac{\sin x}{\sqrt{36 - \cos^2 x}} dx$$

Let ,

$$\cos x = t \Rightarrow -\sin x dx = dt$$

Now, putting

$$\sin x dx = -dt$$

and,

$$\cos x = t$$

$$= \int \frac{dt}{\sqrt{36 - t^2}}$$

$$= -\int \frac{dt}{\sqrt{(6)^2 - t^2}}$$

$$= -\sin^{-1}\left(\frac{t}{6}\right) + c$$

$$= -\sin^{-1}\left(\frac{\cos x}{6}\right) + c \quad [\because t = \cos x]$$

Question 5

[14]

(A) Attempt any Two of the following:

[6]

(i) If $y = f(u)$ is a differentiable function of u and $u = g(x)$ is differentiable function of x then prove that $y = f(g(x))$ is a differentiable function of x and $\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$.

(3)

Sol:

We have,

Let $\delta x, \delta y$ and δu be a small change in x, y and u respectively.

As, $\delta x \rightarrow 0, \delta y \rightarrow 0, \delta u \rightarrow 0$

As u is differentiable function, it is continuous.

Now, consider the increment ratio $\frac{\delta y}{\delta x}$

So, we have,

$$\frac{\delta y}{\delta x} = \frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x}$$

Taking limit as $\delta x \rightarrow 0$ on both sides

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \left(\frac{\delta y}{\delta u} \times \frac{\delta u}{\delta x} \right)$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x}$$

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta u \rightarrow 0} \frac{\delta y}{\delta u} \times \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x} \text{----- (1)}$$

Since, y is a differentiable function of u , $\lim_{\delta u \rightarrow 0} \left(\frac{\delta y}{\delta u} \right)$ exists and $\lim_{\delta x \rightarrow 0} \left(\frac{\delta u}{\delta x} \right)$ exists as

u is a differentiable function of x .

RHS of equation (1) exists.

Now,

$$= \lim_{\delta u \rightarrow 0} \frac{\delta y}{\delta u} = \frac{dy}{du} \text{ and } \lim_{\delta x \rightarrow 0} \frac{\delta u}{\delta x} = \frac{du}{dx}$$

$$\therefore \lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{du} \times \frac{du}{dx}$$

So, RHS exists.

Therefore, LHS also exists and

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Hence, proved.

(ii) The probability that a person who undergoes kidney operation will recover is 0.5. Find the probability that of the six patients who undergo similar operations. (a) None will recover. (b) Half of them will recover.

(3)

Sol:

Given that,

Since there are six patients $n = 6$

If a patient recovers, the outcome is a success.

Thus,

$$P = p(\text{success}) = 0.5 \text{ and } q = 1 - p = 0.5$$

let X : No. of patients, who recovered out of 6.

Then $X \sim B(n = 6, p = 0.5)$

The p.m.f of X is given as,

$$P(X = x) = p(x) = {}^6C_x (0.5)^x (0.5)^{6-x}, x = 0, 1, 2, 3, 4, 5, 6.$$

(i) Probability (none will recover),

$$\begin{aligned} P(X = 0) &= {}^6C_0 (0.5)^0 (0.5)^6 \\ &= (1) \cdot (1) \cdot (0.5)^6 \\ &= 0.015625 \end{aligned}$$

(ii) Probability (Half of them will recover),

$$\begin{aligned} P(X = 3) &= {}^6C_3 (0.5)^3 (0.5)^3 \\ &= 20 \times 0.125 \times 0.125 \\ &= 0.3125 \end{aligned}$$

(iii) Evaluate: $\int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$ (3)

Sol:

We have,

$$I = \int_0^{\pi} \frac{x}{a^2 \cos^2 x + b^2 \sin^2 x} dx \text{----- (1)}$$

$$I = \int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 (\pi - x) + b^2 \sin^2 (\pi - x)} dx \quad (\because \int_0^a f(x) dx = \int_0^a f(a - x) dx)$$

$$I = \int_0^{\pi} \frac{\pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} dx \text{----- (2)}$$

On adding equation (1) and (2), we get

$$2I = \int_0^{\pi} \frac{x + \pi - x}{a^2 \cos^2 x + b^2 \sin^2 x} dx$$

$$2I = \pi \int_0^{\pi} \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx$$

Using the property,

$$\int_0^{2a} f(x).dx = \int_0^{2a} [f(x) + f(2a-x)]dx$$

$$\therefore 2I = \left[\int_0^{\frac{\pi}{2}} \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx + \int_0^{\frac{\pi}{2}} \frac{1}{a^2 \cos^2(\pi-x) + b^2 \sin^2(\pi-x)} dx \right]$$

$$I = \pi \int_0^{\frac{\pi}{2}} \frac{1}{a^2 \cos^2 x + b^2 \sin^2 x} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{dx}{a^2 \cos^2 x \left(1 + \frac{b^2}{a^2} \tan^2 x \right)}$$

$$= \frac{\pi}{a^2} \int_0^{\frac{\pi}{2}} \frac{\sec^2 x dx}{1 + \frac{b^2}{a^2} \tan^2 x} \quad \left[\text{by putting } u = \frac{b}{a} \tan x, du = \frac{b}{a} \sec^2 x dx \text{ when } x=0, u=0, \text{ when } x=\frac{\pi}{2}, u=\infty \right]$$

$$= \frac{\pi}{a^2} \int_0^{\frac{\pi}{2}} \frac{\frac{a}{b} du}{1+u^2}$$

$$= \frac{\pi}{ab} \int_0^{\frac{\pi}{2}} \frac{du}{1+u^2} = \frac{\pi}{ab} [\tan^{-1}(u)]_0^{\infty} + c$$

$$= \frac{\pi}{ab} [\tan^{-1} \infty - \tan^{-1} 0] + c$$

$$= \frac{\pi}{ab} \left[\frac{\pi}{2} - 0 \right] + c$$

$$= \frac{\pi^2}{2ab}$$

(B) Attempt any Two of the following:

[8]

(i) Discuss the continuity of the following functions. If the function have a removable discontinuity, redefine the function so as to remove the discontinuity.

(4)

$$f(x) = \frac{4^x - e^x}{6^x - 1}, \text{ For } x \neq 0 \quad \left. \vphantom{\frac{4^x - e^x}{6^x - 1}} \right\} \text{ as } x \rightarrow 0$$

$$= \log\left(\frac{2}{3}\right), \text{ For } x = 0$$

Sol:

We have,

$$f(0) = \log\left(\frac{2}{3}\right)$$

Now,

$$\begin{aligned}\lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{4^x - e^x}{6^x - 1} \\&= \lim_{x \rightarrow 0} \frac{(4^x - 1) - (e^x - 1)}{6^x - 1} \\&= \lim_{x \rightarrow 0} \left[\frac{(4^x - 1) - (e^x - 1)}{x} \cdot \frac{x}{6^x - 1} \right] \\&= \frac{\lim_{x \rightarrow 0} \left(\frac{4^x - 1}{x} \right) - \lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right)}{\lim_{x \rightarrow 0} \frac{6^x - 1}{x}} \\&= \frac{\log 4 - \log e}{\log 6} = \frac{\log \left(\frac{4}{e} \right)}{\log 6} \\&= \lim_{x \rightarrow 0} f(x) \neq f(0)\end{aligned}$$

$\therefore f(x)$ is discontinuous at $x = 0$

So, $\lim_{x \rightarrow 0} f(x)$ exists, but not equal to $f(0)$. Hence, the discontinuity at $x = 0$ is removable and it can be removed by redefining the function as follows:

$$\begin{aligned}f(x) &= \frac{4^x - e^x}{6^x - 1} \text{ for } x \neq 0 \\&= \frac{\log \left(\frac{4}{e} \right)}{\log 6}, \text{ for } x = 0\end{aligned}$$

(ii) Prove that: $\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c$ (4)

Sol:

We have,

$$\begin{aligned}I &= \int \sqrt{a^2 - x^2} dx \\&= \int \sqrt{a^2 - x^2} \cdot 1 dx\end{aligned}$$

Integrating by parts

$$\begin{aligned}
 I &= \int \sqrt{a^2 - x^2} \int 1 dx - \int \left[\frac{d}{dx} (\sqrt{a^2 - x^2}) \cdot \int 1 dx \right] dx \\
 &= x\sqrt{a^2 - x^2} - \int \frac{-2x}{2\sqrt{a^2 - x^2}} x dx \\
 &= x\sqrt{a^2 - x^2} - \int \frac{(a^2 - x^2) - a^2}{\sqrt{a^2 - x^2}} dx \\
 &= x \cdot \sqrt{a^2 - x^2} - \int \left[\sqrt{a^2 - x^2} - \frac{a^2}{\sqrt{a^2 - x^2}} \right] dx \\
 &= x \cdot \sqrt{a^2 - x^2} - \int \sqrt{a^2 - x^2} dx + a^2 \int \frac{1}{\sqrt{a^2 - x^2}} dx \\
 &= x \cdot \sqrt{a^2 - x^2} - I + a^2 \sin^{-1} \left(\frac{x}{a} \right) + c_1 \\
 2I &= x \cdot \sqrt{a^2 - x^2} + a^2 \sin^{-1} \left(\frac{x}{a} \right) + c_1 \\
 I &= \frac{x}{2} \cdot \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c_1 \\
 \therefore \int \sqrt{a^2 - x^2} dx &= \frac{x}{2} \cdot \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) + c \text{ where, } c = \frac{c_1}{2}
 \end{aligned}$$

Hence, proved.

(iii) A body is heated at 110°C and placed in air at 10°C. After 1 hour its temperature is 60°C. How much additional time is required for it to cool to 35°C?

(4)

Sol:

Let θ be the temperature of body at time t . Temperature of air is given to be 10°C. Then, according to Newton's law of cooling, we have

$$\begin{aligned}
 \frac{d\theta}{dt} &\propto (\theta - 10^0) \\
 \frac{d\theta}{dt} &= -k(\theta - 10^0), k > 0 \\
 \Rightarrow \frac{d\theta}{\theta - 10^0} &= -k dt
 \end{aligned}$$

On integrating both sides, we get

$$\int \frac{d\theta}{\theta - 10^0} = -k \int dt$$

$$\ln(\theta - 10^0) = -kt + c$$

$$\theta = 10^0 + e^{-kt}$$

$$\text{when } t = 0, \theta = 110^0$$

Substituting the above equation, we get

$$110^0 = 10^0 + e^{-k(0)+c}$$

$$e^c = 100^0$$

Substituting the above equation in (1), we get

$$\theta = 10^0 + 100^0 e^{-kt}$$

Given that, $\theta = 60^0 C$ when $t = 1$, we get

$$60^0 = 10^0 + 100^0 e^{-k}$$

$$50^0 = 100^0 e^{-k}$$

$$e^{-k} = \frac{1}{2} \Rightarrow e^k = 2$$

$$k = \log 2$$

Now, we have find t, when $\theta = 35^0$

From equation (2), we have

$$\theta = 10^0 + 100^0 e^{-kt}$$

$$35^0 = 10^0 + 100^0 e^{-kt}$$

$$25^0 = 100^0 e^{-kt}$$

$$e^{-kt} = \frac{1}{4} \Rightarrow e^{kt} = 4$$

$$kt = \log 4$$

$$t = \frac{\log 4}{\log 2}$$

Therefore, Additional time $t = \frac{\log 4}{\log 2} - 1 = 1 \text{ hour}$.

Question 6

[14]

(A) Attempt any Two of the following:

[6]

(i)prove that: $\int_0^{2a} f(x)dx = \int_0^a f(x)dx + \int_0^a f(2a-x)dx$ (3)

Sol:

$$\begin{aligned} R.H.S &= \int_0^a f(x)dx + \int_0^a f(2a-x)dx \\ &= I_1 + I_2 \end{aligned}$$

For,

$$I_2 = \int_0^a f(2a-x)dx$$

Put,

$$2a-x=t$$

$$\therefore dx = -dt$$

$$\text{when } x=0 \text{ then } t=2a$$

$$\text{when } x=a \text{ then } t=a$$

$$\begin{aligned} \int_0^a (2a-x)dx &= \int_{2a}^a f(t)(-dt) \\ &= -\int_{2a}^a f(t)dt \quad \left(\because \int_a^b f(x)dx = -\int_b^a f(x)dx \right) \end{aligned}$$

$$\therefore \int_0^a (2a-x)dx = \int_a^{2a} f(x)dx$$

$$\begin{aligned} \therefore I_1 + I_2 &= \int_0^a f(x)dx + \int_a^{2a} f(x)dx \\ &= \int_0^{2a} f(x)dx \end{aligned}$$

Hence, RHS = LHS proved.

(ii)Evaluate: $\int \frac{1+\log x}{x(2+\log x)(3+\log x)}dx$ (3)

Sol:

Let,

$$I = \int \frac{1+\log x}{x(2+\log x)(3+\log x)}dx$$

Now, put

$$1 + \log x = \frac{1}{x} dx = dt$$

$$I = \int \frac{t}{(1+t)(2+t)} dt \text{-----(1)}$$

Let,

$$\frac{t}{(1+t)(2+t)} = \frac{A}{1+t} + \frac{B}{2+t}$$

$$t = A(2+t) + B(1+t) \text{-----(2)}$$

Putting $t = -1$ in equation (2), we get

$$-1 = A \Rightarrow A = -1$$

Putting $t = -2$ in equation (2), we get

$$-2 = -B \Rightarrow B = 2$$

Then equation (1), becomes

$$\begin{aligned} I &= \int \left[\frac{-1}{1+t} + \frac{2}{2+t} \right] dt \\ &= -\int \frac{1}{1+t} dt + 2 \int \frac{1}{2+t} dt \\ &= -\log|t+1| + 2\log|t+2| + c \\ &= 2\log|\log x + 3| - \log|\log x + 2| + c \\ &= \log|(\log x + 3)^2| - \log|(\log x + 2)| + c \\ \therefore \int \frac{1 + \log x}{x(2 + \log x)(3 + \log x)} dx &= \log \left| \frac{(\log x + 3)^2}{(\log x + 2)} \right| + c \end{aligned}$$

(iii) If $y = \cos^{-1}(2x\sqrt{1-x^2})$, find $\frac{dy}{dx}$ (3)

Sol:

Let,

$$y = \cos^{-1}(2x\sqrt{1-x^2})$$

Put,

$$x = \sin \theta$$

$$\therefore \theta = \sin^{-1} x$$

$$y = \cos^{-1} \left(2 \sin \theta \sqrt{1 - \sin^2 \theta} \right)$$

$$y = \cos^{-1} (2 \sin \theta \cos \theta)$$

$$y = \cos^{-1} (\sin 2\theta)$$

$$y = \cos^{-1} \left[\cos \left(\frac{\pi}{2} - 2\theta \right) \right]$$

$$y = \frac{\pi}{2} - 2\theta$$

$$y = \frac{\pi}{2} - 2 \sin^{-1} x$$

$$\therefore \frac{dy}{dx} = \frac{-2}{\sqrt{1-x^2}}$$

(B) Attempt any Two of the following:

[8]

(i) solve the differential equation $\frac{dy}{dx}$

(4)

Hence find the particular solution for $\frac{dy}{dx}$ and $\frac{dy}{dx}$

Sol:

The given differential equation is

$$\frac{dy}{dx} = \frac{1}{\cos(x+y)}$$

Let,

$$x + y = v$$

$$1 + \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dy}{dx} = \frac{dv}{dx} - 1$$

Substituting $\frac{dy}{dx}$ in equation (1), we get

$$\frac{dv}{dx} - 1 = \frac{1}{\cos v}$$

$$\frac{dv}{dx} = \frac{1}{\cos v} + 1$$

$$\frac{dv}{dx} = \frac{1 + \cos v}{\cos v}$$

Now, Integrating,

$$\int \frac{\cos v dv}{1 + \cos v} = \int dx + c$$

$$\int \frac{1 + \cos v}{1 + \cos v} dv - \int \frac{1}{1 + \cos v} dv = \int dx + c$$

$$\int 1 dv - \int \frac{1}{2 \cos^2 \frac{v}{2}} dv = x + c$$

$$v - \frac{1}{2} \int \sec^2 \frac{v}{2} dv = x + c$$

$$v - \tan \frac{v}{2} = x + c$$

$$(x + y) - \tan \left(\frac{x + y}{2} \right) = x + c$$

Which is the required general solution of the given equation.

For $x = 0$ and $y = 0$ equation (2) becomes,

$$(0 + 0) - \tan \left(\frac{0}{2} \right) = 0 + c$$

$$\Rightarrow c = 0$$

Now, putting the value of c in equation (2), We get

$$x + y - \tan \left(\frac{x + y}{2} \right) = x$$

$$\Rightarrow y = \tan \left(\frac{x + y}{2} \right)$$

(ii) A wire of length I is cut into two parts. One part is bent into a circle and other into a square. Show that the sum of areas of the circle and square is the least, if the radius of circle is half the side of the square.

(4)

Sol:

Given that,

Length of wire is l .

Let length of one part be x , then other part be $l - x$

Let x part be bent into a circle and $l - x$ part be in a square.

$$\therefore 2\pi r = x \text{ and } 4s = l - x \Rightarrow s = \frac{l - x}{4} \quad (\because s = \text{side})$$

According to question,

$$r = \frac{s}{2} = \frac{l - x}{8}$$

Also,

$$\begin{aligned} r &= \frac{x}{2\pi} \\ \Rightarrow \frac{l - x}{8} &= \frac{x}{2\pi} \\ \Rightarrow l &= \frac{8x}{2\pi} + x = \frac{4x}{\pi} + x \end{aligned}$$

Sum of areas of circle and square,

$$\begin{aligned} f(x) &= \pi \left(\frac{l - x}{8} \right)^2 + \left(\frac{l - x}{4} \right)^2 \\ f(x) &= \frac{\pi}{64} [(l - x)^2 + 4(l - x)^2] \\ f(x) &= \frac{\pi}{64} 5(l - x)^2 \\ &= \frac{5\pi}{64} \left(\frac{4x}{\pi} + x - x \right)^2 \\ f(x) &= \frac{5\pi}{64} \frac{16x^2}{\pi^2} \\ f(x) &= \frac{5x^2}{4\pi} \\ \therefore f'(x) &= \frac{5x^2}{4\pi} \cdot 2x = \frac{5x}{2\pi} \end{aligned}$$

For minimum value,

$$f'(x) = 0$$

$$\frac{5}{2\pi}x = 0$$

$$x = 0$$

$$\text{If } x = 0, f''(0) = \frac{5}{2\pi} > 0$$

$\therefore f(x)$ attains minimum at $x = 0$.

(iii) The following is the p.d.f.(probability Density Function) of X: (4)

$$f(x) = \frac{x}{32}, 0 < x < 8$$

$= 0, \text{ otherwise}$

(a) Find the expression for c.d.f (Cumulative Distribution Function) of X.

Sol:

c.d.f of a continuous random variable X is given by

$$f(x) = \int_{-\infty}^x f(y)dy$$

$$\therefore f(x) = \int_0^x f(y)dy = \int_0^x \frac{y}{32} dy$$

$$= \left[\frac{y^2}{64} \right]_0^x = \frac{x^2}{64}$$

Thus,

$$F(x) = \frac{x^2}{64}, x \in R$$

(b) Also find its value at $x = 0.5$ and 9.

Sol:

Values of $F(x)$ at different values of x .

$$\text{At } x = 0.5, F(x) = F(0.5) = \frac{(0.5)^2}{64} = \frac{0.25}{64} = \frac{1}{256}$$

For any value of x greater than or equal to 8, $F(x) = 1$

$$\therefore F(9) = 1$$

